Legacy Database Information

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Contents

Note

This page contains information about legacy Kaptive databases for backwards compatibility and interpretation of old results.

Legacy *Klebsiella* O locus database

The *Klebsiella* O locus database (`Klebsiella_o_locus_primary_reference.gbk`) contains annotated sequences for 13 distinct *Klebsiella* O loci.

O locus classification requires some special logic, as the O1 and O2 serotypes are associated with the same loci and the distinction between O1 and each of the four defined O2 subtypes (O2a, O2afg, O2ac, O2aeh) is determined by the presence/absence of 'extra genes' elsewhere in the chromosome as indicated in the table below. Kaptive therefore looks for these genes to predict antigen (sub)types. (Note that the original implementation of O locus typing in Kaptive (< v2.0) distinguished O1 and O2 but not the O2 subtypes.)

Read more about the O locus and its classification here: The diversity of *Klebsiella* pneumoniae surface polysaccharides.

Find out about the genetic determinants of O1 and O2 (sub)types here: Molecular basis for the structural diversity in serogroup O2-antigen polysaccharides in *Klebsiella pneumoniae*.

Find out about the O1 glycoforms and their genetic determinants here: Identification of a second glycoform of the clinically prevalent O1 antigen from *Klebsiella pneumoniae*.

Database versions:

stable

- Kaptive v0.4.0 and above include the original version of the *Klebsiella* O locus database, as described in Wick, R. et al. J Clin Microbiol 2019.
- Kaptive v2.0 and above include a novel O locus reference (O1/O2v3) and updated ‘Extra genes’ for prediction of O1 and O2 antigen (sub)types, as shown in the table below and described in Lam, M.M.C et al. 2021. Microbial Genomics 2022.
- Kaptive v2.0.8 and above include:
    i. updated ‘Extra genes’ logic for prediction of O1 glycoforms, reported as O1a (isolate predicted to produce O1a only) and O1ab (isolate predicted to be able to produce both O1a and O1b glycoforms);
    ii. OL101 re-assigned as OL13 and its associated phenotype prediction updated to O13, to reflect the description of the novel O13 polysaccharide structure.

Genetic determinants of O1 and O2 outer LPS antigens as reported in Kaptive:

stable

O locus	Extra genes	Kaptive < v2.0 (locus[a])	Kaptive v2.0+ (locus[a])	Kaptive v2.0 - v2.0.7 (type[b])	Kaptive v2.0.8+ (type[b])
O1/O2v1	none	O2v1	O1/O2v1	O2a	O2a
O1/O2v2	none	O2v2	O1/O2v2	O2afg	O2afg
O1/O2v3	none	Na	O1/O2v3	O2a	O2a
O1/O2v1	*wbbYZ*	O1v1	O1/O2v1	Na	O1ab
O1/O2v2	*wbbYZ*	O1v2	O1/O2v2	Na	O1ab
O1/O2v3	*wbbYZ*.	Na	O1/O2v3	Na	O1ab
O1/O2v1	*wbbY* only	O1v1	O1/O2v1	O1	O1a
O1/O2v2	*wbbY* only	O1v2	O1/O2v2	O1	O1a
O1/O2v3	*wbbY* only	Na	O1/O2v3	O1	O1a
O1/O2v1	*wbbY* OR *wbbZ*	O1/O2v1	Na	Na	Na
O1/O2v2	*wbbY* OR *wbbZ*	O1/O2v2	Na	Na	Na
O1/O2v3	*wbbY* OR *wbbZ*	Na	Na	Na	Na
O1/O2v1	*wbmVW*	Na	O1/O2v1	O2ac	O2ac
O1/O2v2	*wbmVW*	Na	O1/O2v2	O2ac	O2ac
O1/O2v3	*wbmVW*	Na	O1/O2v3	O2ac	O2ac
O1/O2v1	*gmlABD*	Na	O1/O2v1	O2aeh	O2aeh
O1/O2v2	*gmlABD*	Na	O1/O2v2	O2aeh	O2aeh
O1/O2v3	*gmlABD*	Na	O1/O2v3	O2aeh	O2aeh
O1/O2v1	*wbbY* AND *wbmVW*	Na	O1/O2v1	O1 (O2ac)	

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O locus	Extra genes	Kaptive < v2.0 (locus[a])	Kaptive v2.0+ (locus[a])	Kaptive v2.0 - v2.0.7 (type[b])	Kaptive v2.0.8+ (type[b])
O1/O2v2	*wbbY* AND *wbmVW*	Na	O1/O2v2	O1 (O2ac)[b]	O1 (O2ac)[b]
O1/O2v3	*wbbY* AND *wbmVW*	Na	O1/O2v3	O1 (O2ac)[b]	O1 (O2ac)[b]

- [a] as reported in the ‘Best match locus’ column in the Kaptive output.
- [b] predicted antigenic serotype reported in the ‘Best match type’ column in the Kaptive output (v2.0 and above).
- Na- not applicable

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